

Factorial Holdout Suite Runbook

This suite implements the supervisor-requested holdout-first factorial ablation in the original resource-collection environment.

Goal

Compare six curriculum recipes while treating held-out opponent win rate as the primary endpoint:

1. fixed_predator_holdout_endpoint
2. rotating_opponents_holdout_endpoint
3. rotating_plus_nemesis_archive
4. rotating_plus_novelty_gate
5. rotating_plus_replay_aware_selection
6. rotating_plus_nemesis_novelty_replay

Training score remains useful, but the main ranking criterion is held-out opponent win rate after training.

Replication Target

Use five replicate seed offsets. That yields 6 conditions x 5 replicated suite runs = 30 condition-runs, which matches the supervisor target of 20-30 seeds.

Recommended labels and offsets:

- a: 0
- b: 1000
- c: 2000
- d: 3000
- e: 4000

Commands

```
python run_suite.py --config configs\factorial_holdout_suite\01_factorial_holdout.json --seed-offset 0
--replicate-label a
python run_suite.py --config configs\factorial_holdout_suite\01_factorial_holdout.json --seed-offset 1000
--replicate-label b
python run_suite.py --config configs\factorial_holdout_suite\01_factorial_holdout.json --seed-offset 2000
--replicate-label c
python run_suite.py --config configs\factorial_holdout_suite\01_factorial_holdout.json --seed-offset 3000
--replicate-label d
python run_suite.py --config configs\factorial_holdout_suite\01_factorial_holdout.json --seed-offset 4000
--replicate-label e
```

Then aggregate:

```
python aggregate_runs.py --runs-root runs\factorial_holdout_suite
```

Interpretation Order

1. Rank conditions by held-out win rate first.

2. Use held-out score margin as a secondary endpoint.
3. Use training score only as supporting context.
4. Use novelty, replay robustness, and curriculum metrics to explain why a condition won or lost.
5. Do not treat code novelty alone as innovation until the novelty-review packet confirms real behavioral change.

Follow-Up

After the aggregate is produced, run the novelty-review tooling on the top novelty spikes before making strong claims about innovation:

```
python review_novelty_spikes.py --runs-root runs\factorial_holdout_suite --output-dir  
runs\factorial_holdout_suite\novelty_review
```